

ANISOTROPIC COMPTONIZATION IN THERMAL PLASMAS: SPECTRAL DISTRIBUTION IN PLANE-PARALLEL GEOMETRY

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ABSTRACT

We present a semi-analytical formulation of the problem of inverse Compton scattering of low-frequency radiation in a hot thermal plasma. The scattering medium is assumed to have a plane-parallel geometry, and an input radiation field is localized on one side of the slab. The angular and energy distributions of the emitted radiation are computed taking into account the effects due to the anisotropy of the input photons. We show how the Comptonized spectrum depends on three parameters, namely the optical depth of the medium, the temperature of thermal electrons, and the viewing angle. The spectral shape is approximately, but not exactly, a broken power law. Our method allows to compute angle-dependent Compton spectra for a broad range of input parameters with a modest computational effort.

Subject headings: radiation mechanisms: miscellaneous — radiative transfer

1. INTRODUCTION

Inverse Compton (IC) emission may play an important role in producing high-energy photons in compact sources. Radiative models for radio-quiet AGNs (e.g., Czerny & Elvis 1987; Zdziarski et al. 1990; Haardt & Maraschi 1991) invoke this effect as the main X-ray and γ -ray production mechanism. In these models a low-frequency radiation field (soft photon input) is Comptonized by hot electrons. The input photons are thermally emitted by nearby cold matter, possibly an accretion flow. In AGNs there is evidence that the accretion flow takes place in a disk fashion. Hot electrons may form an active corona located above the surface of the disk. The presence of a corona surrounding an accretion disk is suggested also in Galactic X-ray binaries (Ebisawa 1990). In this model, the source of soft photons is not homogeneously distributed within the scattering medium, so that the first scattering event is highly anisotropic. Many papers have been devoted to the study of Comptonization of low-energy radiation in hot plasmas (e.g., Sunyaev & Titarchuk 1980; Lightman & Rybicki 1980). Nevertheless only recently Ghisellini et al. (1991), Rogers & Field (1991), and Haardt & Maraschi (1991) suggested the importance of the anisotropic inverse Compton process in AGNs. In these works it has mainly been emphasized that the total inverse Compton power emitted in different directions depends on direction itself. This has important consequences for the fluorescent line emission observed in Seyfert galaxies (Nandra et al. 1989; Pounds et al. 1990), and for the X-ray background model proposed by Fabian et al. (1990). Ghisellini et al. (1991) and Rogers & Field (1991) assumed a nonthermal distribution of ultrarelativistic electrons. In this case the photon spectrum is essentially formed by a single scattering process.

In thermal media the situation is different, since even for low optical depths the spectrum is shaped by multiple scatterings. Since only the first scatterings keep memory of the anisotropy of the source photons, spectral distortions are therefore present in the low-energy part of the Comptonized radiation field. This has been shown in Haardt & Maraschi (1991), where anisotropic IC is an essential feature of the model.

In the present paper we devise a formal method to derive the spectral shape in the case of a plane parallel scattering plasma. The novelty of the method is that the angular dependence of the emitted spectrum is taken into account, together with the energy distribution. Moreover, no limitation on the optical depth is required. Our calculations are formally developed for thermal electrons since in this case multiple scatterings play an important role. The generality of the method allows one to consider different electron energy distributions. Our solutions neglect electron recoil, and are in good agreement with Monte Carlo simulations as long as the Thomson approximation is valid in the electron rest frame. This means that our method is correct for $h\nu \ll \min(kT, m_e c^2/\langle\gamma\rangle)$, where m_e and $\langle\gamma\rangle$ are the rest mass and the average Lorentz factor of the electrons, respectively. In the accompanying paper (Haardt & Maraschi 1993) the method is applied in order to treat Comptonization in the two phase model (Haardt & Maraschi 1991). Our calculations have also interest in view of estimating the X-ray polarization in AGNs and X-ray binaries (Haardt & Matt 1993).

Work along the same line has been carried out by Nishimura, Mitsuda, & Itoh (1986). In their work the angular dependence of the emitted radiation is not taken into account. The authors consider isotropic scattering in the laboratory frame, limiting the validity of the results to $kT < 150$ keV.

The paper is structured as follows. In § 2 the calculation method is outlined, while the formalism in the case of anisotropic IC emission is presented in § 3. In § 4 we show some examples of calculation and we discuss the results. The last part of the paper is devoted to conclusions.

2. COMPUTATIONAL METHOD

2.1. General Outline

The spectrum emitted from a Comptonizing atmosphere with optical depth τ can be written as an infinite sum of the single-scattering spectra f_k over the total number of scatterings:

$$f(x) = \sum_{k=0}^{\infty} f_k(x), \quad (1)$$

where x is the photon energy in appropriate units. As discussed in Lightman & Rybicki (1980) and in Nishimura et al. (1986), the spectrum of k -fold scattered photons can be written as a product of two terms:

$$f_k(x) = p_k(\tau) S_k(x). \quad (2)$$

The first term p_k is the escape probability, i.e., the probability of a photon to escape from the source after it has suffered k scatterings. The term S_k can be seen as the photon spectrum in an *infinite* scattering medium. More formally, S_k is the probability density distribution in the three-dimensional space of the photon momenta. A point in this space can be represented as (x, Ω) where Ω defines a unit vector parallel to the photon momentum. The differential quantity $S(x, \Omega) dx d\Omega$ is the probability that a photon with energy x within dx travels within a solid angle $d\Omega$ in direction Ω . By definition

$$\int dx d\Omega S_k(x, \Omega) = 1. \quad (3)$$

Equation (2) states that the photon spectrum emerging from a *finite* medium can be written as the probability density distribution weighted over the escape probability. The advantage of the present formulation is that all the information about the geometry and the opacity of the source is into the $p_k(\tau)$ term. One can deal with two separate problems:

1. The first is to compute the evolution from S_k to S_{k+1} .
2. The second is to compute the probabilities p_k for every k .

The first problem can be solved iteratively by means of the electron emissivity due to inverse Compton (IC) scattering on monoenergetic photons, and can be calculated assuming an isotropic phase function in the electron rest frame, as discussed by Rybicki & Lightman (1979). The *kernel* of single scattering is defined as the electron emissivity averaged over the actual electron energy distribution. The probability density distribution S_{k+1} is the convolution of the kernel with S_k .

The second problem can be solved assuming isotropic scattering in the laboratory frame. Under the above assumptions, and in the case of anisotropic photon input, the angular dependence of the radiation field cannot be however adequately calculated. In order to overcome the problem we thus introduce dimensionless coefficients describing the degree of anisotropy of the scattering. These coefficients are used to compute *effective* values of the electron temperature, and *effective* angular patterns of the scattering. The scattering in the laboratory frame is then modeled by means of an effective phase function, while the energy transfer is computed in the isotropic approximation, considering an electron plasma at a given effective temperature. The power of the method consists in computing the Comptonized spectrum by means of two energy-angle dependent coefficients. The calculation is limited to the Thomson regime in the electron rest frame.

2.2. Evolution of the Probability Density Distribution

In the following we will assume that the scattering medium consists of free electrons isotropically distributed. Due to scattering events the probability density distribution S evolves in the space of photon momenta. We can follow this evolution step by step by means of the kernel Ψ defined by the following iterative equation:

$$S_{k+1}(x, \Omega) = \int dy d\Omega' S_k(y, \Omega') \Psi(x, y, \Omega, \Omega'). \quad (4)$$

The kernel Ψ represents the probability that a photon of momentum (y, Ω') is scattered in a single event into (x, Ω) . In order to solve the iterative integral equation given above the initial condition S_0 must be specified.

Equation (4) can be practically solved only in a few simple cases. For example we can suppose that the photons are distributed isotropically in the Euclidean space. We refer to this condition as *complete isotropy*. Since there is no evolution in Ω , i.e., the isotropy is conserved, the evolution of the probability density distribution involves only the energy component. Equation (4) takes a simplified form:

$$S_{k+1}(x) = \int dy S_k(y) \Psi(x, y). \quad (5)$$

2.3. Kernel for a Maxwellian Electron Distribution

The isotropic kernel $\Psi(x, y)$ is related to the emissivity of an isotropic distribution of photons due to scattering on an isotropic distribution of electrons. Let the initial photon energy and the scattered photon energy be y and x , respectively. In the case of complete isotropy, the emissivity due to a single scattering is (see Rybicki & Lightman 1979):

$$j(x, y, y) = \frac{1}{4y(\gamma^2 - 1)} \begin{cases} (1 + \beta)\epsilon - (1 - \beta) & \frac{1 - \beta}{1 + \beta} < \epsilon < 1 \\ (1 + \beta) - \epsilon(1 - \beta) & 1 < \epsilon < \frac{1 + \beta}{1 - \beta} \\ 0 & \text{otherwise} \end{cases}, \quad (6)$$

where we have assumed monoenergetic electrons with Lorentz factor γ . We have defined $\epsilon = x/y$. The electron velocity (in units of the speed of the light) is $\beta = \sqrt{\gamma^2 - 1}/\gamma$. It may be easily checked that

$$\int_0^\infty dx j(x) = 1 \quad (7a)$$

and

$$\int_0^\infty dx (x - y) j(x) = \frac{4}{3}(\gamma^2 - 1)y. \quad (7b)$$

The result expressed by equation (6) can be obtained assuming an isotropic phase function for the scattering. The use of the more correct Ryleigh phase function gives negligible corrections. Electron recoil has been neglected.

If the scattering electrons are not monoenergetic, the emissivity j has to be averaged over the actual electron energy distribution $N(\gamma)$. The isotropic kernel Ψ is then

$$\Psi(x, y) = \int_{\bar{\gamma}}^\infty d\gamma j(x, y, \gamma) N(\gamma). \quad (8a)$$

The lower integration limit can be easily derived from equation (6):

$$\bar{\gamma} = (1 + \epsilon)/\sqrt{4\epsilon}. \quad (8b)$$

If we consider an isotropic distribution of Maxwellian electrons, the number density per unit energy is

$$N(\gamma) = \frac{1}{\Theta K_2(1/\Theta)} \gamma(\gamma^2 - 1)^{1/2} e^{-\gamma/\Theta}, \quad (9)$$

where Θ is the thermal temperature in unit of the electron mass at rest, and K_2 is the second-order modified Bessel function.

It is easy to see that the total number of electrons is normalized so as

$$\int_1^\infty d\gamma N(\gamma) = 1, \quad (10a)$$

while the fractional mean energy gain per scattering suffered by photons is proportional to

$$\int_1^\infty d\gamma (\gamma^2 - 1) N(\gamma) = \frac{12\Theta^2 K_2(1/\Theta) + 3\Theta K_1(1/\Theta)}{2\Theta K_1(1/\Theta) + K_0(1/\Theta)}. \quad (10b)$$

Under the assumption of complete isotropy we just need to specify the initial condition $S_0(x)$, and equation (5) can then be solved iteratively.

3. ANISOTROPIC INVERSE COMPTON EMISSION

In the general case, the assumption of complete isotropy is unrealistic, because of the possible presence of an anisotropic soft photon input. In this section we drop the assumption of isotropy of the photon distribution; in this case the kernel depends on the angular variables.

Contrary to the case of complete isotropy, in which the fraction dN of scattered photons per unit solid angle and their mean energy $\langle x \rangle$ are independent on direction, anisotropy implies an explicit dependence of dN and $\langle x \rangle$ on Ω . In the scattering process the electron radiative losses are larger (in the laboratory frame) in the backward direction of the input photons, and their magnitude depends on the electron energy (for a complete discussion see Ghisellini et al. 1991). It is important to point out that when the emission is integrated over the whole solid angle, we obtain exactly the same results as in the case of complete isotropy.

3.1. Anisotropy Coefficients

In order to take the anisotropy effects into account, we consider a radiation beam. We can introduce two dimensionless coefficients, $\phi(\omega)$ and $\chi(\omega)$, representing the degree of anisotropy in the scattering event as function of the cosine of the scattering angle ω . The coefficients are defined by the following relations:

$$dN = (1/4\pi) \phi(\omega) d\omega d\varphi \quad (11)$$

and

$$\langle x \rangle = [4/3(\gamma^2 - 1) + 1] \chi(\omega) y. \quad (12)$$

By definition

$$\frac{1}{2} \int_{-1}^1 d\omega \phi(\omega) = 1 \quad (13a)$$

and, defining $\xi(\omega) = \phi(\omega)\chi(\omega)$

$$\frac{1}{2} \int_{-1}^1 d\omega \xi(\omega) = 1. \quad (13b)$$

Besides the scattering angle, the two coefficients depend on Θ .

As discussed in the § 2.1, our strategy consists in modeling the scattering considering populations of electrons at different *effective* temperatures according to the viewing angle. Our basic assumption is that the emissivity due to single scattering has the same form as given by equation (6), but photons are scattered in different directions by electrons with different temperatures, so that their mean energy after the scattering is described by equation (12). Since we are considering the direction of the unscattered radiation beam as reference axis, the kernel $\Psi(x, y, \Omega, \Omega')$ is $\Psi(x, y, \omega, 1)$, which can be rewritten as $\phi(\omega)\tilde{\Psi}(x, y, \omega)$.

From equation (12) one can define an *effective* Lorentz factor of the electrons responsible to the emission:

$$\bar{\gamma}(\omega, \gamma) = \sqrt{[(4\gamma^2 - 1)\chi + 1]/4}. \quad (14)$$

In our case, since we are considering a Maxwellian distribution of electrons, using equation (10b) we define an (angle-dependent) *effective* temperature $\tilde{\Theta}(\omega, \Theta)$ for the scattering as solution of the following equation:

$$\left[\frac{16\Theta^2 K_2(1/\Theta) + 4\Theta K_1(1/\Theta)}{2\Theta K_1(1/\Theta) + K_0(1/\Theta)} + 1 \right] \chi(\omega) = \frac{16\tilde{\Theta}^2 K_2(1/\tilde{\Theta}) + 4\tilde{\Theta} K_1(1/\tilde{\Theta})}{2\tilde{\Theta} K_1(1/\tilde{\Theta}) + K_0(1/\tilde{\Theta})} + 1. \quad (15)$$

Now the (angle-dependent) *effective* number density

$$\tilde{N}(\gamma, \omega) = \frac{1}{\tilde{\Theta} K_2(1/\tilde{\Theta})} \gamma(\gamma^2 - 1)^{1/2} e^{-\gamma/\tilde{\Theta}} \quad (16)$$

can be used to write the modified kernel as

$$\tilde{\Psi}(x, y, \omega) = \int_{\bar{\gamma}}^{\infty} d\gamma j(x, y, \gamma) \tilde{N}(\gamma, \omega). \quad (17)$$

The probability density distribution after the scattering can be calculated by means of a modification of equation (4), performing a single integral over energy:

$$\tilde{S}_{k+1}(x, \omega) = \tilde{S}_{k+1}(x, \phi, \xi) = \phi(\omega) \int dy S_k(y, 1) \tilde{\Psi}(x, y, \omega). \quad (18)$$

In the above calculations a beam of radiation scattered by an isotropic electron distribution has been assumed, considering the direction of the beam as reference axis. In the next § 3.2 these results will be used to model the anisotropic scattering for a generic axisymmetric distribution of input photons, and the probability density distribution $\tilde{S}_{k+1}(x, \Omega)$ will be computed relative to the symmetry axis of the input radiation distribution.

3.2. Anisotropic First Scattering Distribution

Hereafter we consider a given axisymmetric distribution of photons S_0 located on one face of a plane parallel scattering medium. In the assumed geometry the anisotropy of the radiation field is forced to produce relevant spectral distortions in the first scattering distribution only (see §§ 3.4 and 4). In the present section we consider only the evolution from S_0 and $\tilde{S}_1$, although the formalism developed in the previous § 3.1 has a general validity.

In the following we indicate with μ' and μ the cosine of the angles between the reference axis (i.e., the normal at the slab) and the photon direction before and after the scattering, respectively. For the sake of simplicity, we consider a case in which the input photon distribution $S_0(x, \mu)$ can be explicitly written as $S_0(x)U_0(\mu)$. This assumption is appropriate for the first scattering event in many relevant cases.

As a first step, we compute the distribution $P(\omega, \mu, \mu')$ of the scattering angle in direction μ for a given initial direction μ' . The problem can be solved by means of simple geometrical arguments, noting that

$$\omega = \mu\mu' + \sqrt{1 - \mu^2}\sqrt{1 - \mu'^2} \cos(\rho - \rho'), \quad (19)$$

where ρ' and ρ define the azimuthal angle before and after the scattering, respectively. Due to symmetry, we can set $\rho' = 0$, and we find

$$d\rho = \pm(1 - \mu^2 - \mu'^2 + 2\mu\mu'\omega - \omega^2)^{-1/2} d\omega. \quad (20)$$

It is easy to see that

$$P(\omega, \mu, \mu') = 2 \left| \frac{d\rho}{d\omega} \right|, \quad (21)$$

where the factor 2 takes into account the change of sign in $d\rho$. Because of the symmetry of the problem, the distribution P is independent on the azimuthal angle. Using equation (18), we first compute

$$\tilde{S}_1(x, \omega) = \phi(\omega) \int dy S_0(y) \tilde{\Psi}(x, y, \omega) \quad (22)$$

and then the distribution of photon injected along μ' and scattered in direction μ

$$\tilde{S}_1(x, \mu, \mu') = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} d\omega P(\omega, \mu, \mu') \tilde{S}_1(x, \omega), \quad (23a)$$

where the two integration limits read

$$\begin{aligned} \omega_1 &= \mu\mu' - \sqrt{1-\mu^2}\sqrt{1-\mu'^2} \\ \omega_2 &= \mu\mu' + \sqrt{1-\mu^2}\sqrt{1-\mu'^2}. \end{aligned} \quad (23b)$$

Note that

$$\frac{1}{2\pi} \int_{\omega_1}^{\omega_2} d\omega P(\omega, \mu, \mu') = 1. \quad (24)$$

Finally the probability density distribution $\tilde{S}_1(x, \mu)$ for a generic axisymmetric distribution of input photon $U_0(\mu')$ defined from μ'_m and μ'_M is given by

$$\tilde{S}_1(x, \mu) = \int_{\mu'_m}^{\mu'_M} d\mu' U_0(\mu') \tilde{S}_1(x, \mu, \mu') \left/ \int_{\mu'_m}^{\mu'_M} d\mu' U_0(\mu') \right. . \quad (25)$$

Note that $\tilde{S}_1(x, \mu)$ does not depend on the geometry and opacity of the source since it is defined in the space of photon momenta. Since a plane parallel scattering medium has a *finite* vertical extension, photons travelling at grazing angles ($\mu' \sim 0$) have a mean scattering probability higher than photons traveling in the slab along the normal ($\mu' \sim 1$). In order to take this effect into account, the angular distribution $U_0(\mu')$ must be weighted by means of the appropriate source function. Since the direct escape probability for input photon injected along direction μ' is $\propto \exp(-\tau/\mu')$, where τ is the optical depth of the slab calculated along the symmetry axis, in equation (25) we can replace U_0 with a *weighted* photon distribution defined as

$$U(\mu') = [1 - \exp(-\tau/\mu')] U_0(\mu'). \quad (26)$$

The above procedure is valid only for the first-order scattering $\tilde{S}_1$. Higher order scatterings can be treated as in the case of complete isotropy. An average isotropic distribution

$$S_1(x) = \int dy S_0(y) \Psi(x, y) \quad (27)$$

can be used to compute $S_2(x)$ via equation (5).

Unfortunately the integrals in equations (22), (23a), and (25) can be solved only by means of numerical techniques. Note that $P(\omega, \mu, \mu')$ has an integrable pole in the two integration limits (23b).

In order to overcome the problem, we consider averaged coefficients $\phi(\mu, \mu')$ and $\xi(\mu, \mu')$ given by

$$\phi(\mu, \mu') = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} d\omega P(\omega, \mu, \mu') \phi(\omega) \quad (28a)$$

and

$$\xi(\mu, \mu') = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} d\omega P(\omega, \mu, \mu') \xi(\omega). \quad (28b)$$

Note that the correct value of χ in equation (15) is defined as $\xi(\mu, \mu')/\phi(\mu, \mu')$, and not as a mean value computed via equation (28a) or equation (28b). From equation (18) we can calculate the probability density distribution $\tilde{S}_1(x, \phi, \xi) = \tilde{S}_1(x, \mu, \mu')$ as

$$\tilde{S}_1(x, \mu, \mu') = \phi(\mu, \mu') \int dy S_0(y) \tilde{\Psi}(x, y, \mu, \mu') \quad (29)$$

and then use equation (25) to obtain $\tilde{S}_1(x, \mu)$.

The approximated formulation given above is particularly convenient since the integrals in equations (28a) and (28b) can be explicitly solved using for $\phi(\omega)$ and $\xi(\omega)$ a polynomial approximation of arbitrary order. The calculation is described in § 3.3 and is used in the computation of the IC spectra shown in § 4.

Since for $\mu = 1$ $P(\omega, 1, \mu')$ reduces to a delta function, equation (23a) and equation (29) give exactly the same result. For $\mu = 0$, the distribution $P(\omega, 0, \mu')$ is as broad as possible for every μ' . The use of equation (29) instead of equation (23a) introduces an error increasing with increasing inclination angle.

A further approximation consists in computing an average value of $\phi(\mu, \mu')$ and $\xi(\mu, \mu')$, i.e.,

$$\phi_1(\mu) = \int_{\mu'_m}^{\mu'_M} d\mu' \phi(\mu, \mu') U(\mu') \left/ \int_{\mu'_m}^{\mu'_M} d\mu' U(\mu') \right. \quad (30a)$$

and

$$\xi_1(\mu) = \int_{\mu'_m}^{\mu'_M} d\mu' \xi(\mu, \mu') U(\mu') \left/ \int_{\mu'_m}^{\mu'_M} d\mu' U(\mu') \right., \quad (30b)$$

and then writing directly the distribution $\tilde{S}_1$ using equation (18):

$$\tilde{S}_1(x, \mu) = \tilde{S}(x, \phi_1, \xi_1). \quad (31)$$

As seen for the first average calculation, the coefficient $\chi_1(\mu)$ in equation (15) must be written as $\xi_1(\mu)/\phi_1(\mu)$. The above equation (31) is useful since it gives a good estimate of $\tilde{S}_1(x, \mu)$ in a simple way.

3.3. Polynomial Approximations of $\phi(\omega)$ and $\xi(\omega)$

The computational method previously described needs a tractable form for $\phi(\omega)$ and $\xi(\omega)$. We recall that the two coefficients are functions of the electron temperature, besides the scattering angle. The most convenient procedure is to find polynomial approximations for $\phi(\omega)$ and $\xi(\omega)$. Integrals in equation (28a) and (28b), or in equation (30a) and (30b) can then be solved, and the effective temperature Θ obtained from equation (15). We note however that for high Θ and $\omega \simeq 1$ the anisotropy coefficients tend to be ~ 0 (see below), and the ratio of the two polynomials could diverge when both the coefficients are small. It is then convenient to find an analytical approximation for $\chi(\omega)$, writing $\xi(\omega)$ as $\chi(\omega)\phi(\omega)$.

We have performed Monte Carlo simulations for single scattering for a beam of monoenergetic radiation, computing $\phi(\omega)$ and $\chi(\omega)$ in 20 angular bins with width $\Delta(\omega) = 0.1$, for eight different values of Θ (0.05, 0.1, 0.25, 0.5, 0.75, 1., 3., 5.). Then we have performed a two-dimensional fitting to simulations by means of third-order polynomials in ω :

$$\phi(\omega) = \sum_{i=0}^3 Q_i(\Theta)\omega^i \quad (32)$$

and

$$\chi(\omega) = \sum_{i=0}^3 R_i(\Theta)\omega^i. \quad (33)$$

From $\xi = \phi\chi$, using the above expressions we find

$$\xi(\omega) = \sum_{i=0}^6 Z_i(\Theta)\omega^i, \quad (34)$$

where the coefficients Z can be expressed as combinations of Q and R .

From equations (13a) and (13b) of § 3.1 we have a constrain for Q_0 and Z_0 :

$$Q_0 = 1 - \frac{Q_2}{3} \quad (35a)$$

and

$$Z_0 = 1 - \frac{Z_6}{7} - \frac{Z_4}{5} - \frac{Z_2}{3}. \quad (35b)$$

By definition (eq. [11]), in the limit of low-temperature, $\phi(\omega)$ coincides with the Ryleigh phase function time 4π , while (eq. [12]) $\chi(\omega)$ is $\simeq 1$. A further constrain is due to the fact that for high temperatures the effects of anisotropy do saturate (Ghisellini et al. 1991). Then we write the Q 's in the following suitable form:

$$\begin{aligned} Q_3(\Theta) &= \frac{c_{32}\Theta^2 + c_{31}\Theta}{16\Theta^2 + 4\Theta + 1} \\ Q_2(\Theta) &= \frac{c_{22}\Theta^2 + c_{21}\Theta + 3/4}{16\Theta^2 + 4\Theta + 1} \\ Q_1(\Theta) &= \frac{c_{12}\Theta^2 + c_{11}\Theta}{16\Theta^2 + 4\Theta + 1} \end{aligned} \quad (36a)$$

and the R 's as

$$\begin{aligned} R_3(\Theta) &= \frac{c_{32}\Theta^2 + c_{31}\Theta}{16\Theta^2 + 4\Theta + 1} \\ R_2(\Theta) &= \frac{c_{22}\Theta^2 + c_{21}\Theta}{16\Theta^2 + 4\Theta + 1} \\ R_1(\Theta) &= \frac{c_{12}\Theta^2 + c_{11}\Theta}{16\Theta^2 + 4\Theta + 1}. \end{aligned} \quad (36b)$$

The best-fit values of the numerical coefficients c relative to the ϕ and χ fitting are reported in Table 1. In Figure 1 the polynomial approximation is compared with numerical results. The fits are sufficiently good for our purposes.

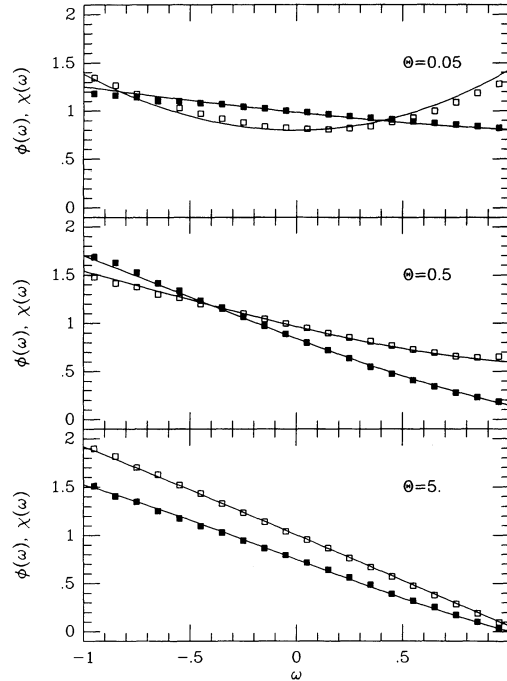


FIG. 1.—Monte Carlo calculations and polynomial approximations of the anisotropy coefficients $\phi(\omega)$ and $\chi(\omega)$ vs. the cosine of the scattering angle ω . From top to the bottom the electron temperature is $\Theta = 0.05$, $\Theta = 0.5$, and $\Theta = 5$, respectively. Open squares refer to ϕ , filled squares refer to χ , as computed via numerical simulations. The polynomial approximations discussed in § 3.3 are shown as continuous lines.

As previously discussed we have used the mean values defined in equations (28a) and (28b) in most of our computations. We remind that, in order to solve equation (15), the proper value of $\chi(\mu, \mu')$ should be computed as $\xi(\mu, \mu')/\phi(\mu, \mu')$. The integrations in equations (28a) and (28b) are cumbersome but not difficult, and results are reported in the Appendix. Results of the second-step averaging (eqs. [30a] and [30b]) depend on the actual angular distribution of the input photons. In the next § 3.4 we will discuss this problem, giving (see next eq. [39]) a generic expression for U_0 in the case of thermal emission from accretion disk. The results of integrations are reported in the Appendix, where the exponential integral functions $E_n(z)$ are defined as

$$E_n(z) = \int_0^1 d\mu \exp(-z/\mu) \mu^{n-2} \quad (37a)$$

and they satisfy the following iterative relation:

$$E_{n+1}(z) = \frac{1}{n} [e^{-z} - zE_n(z)] . \quad (37b)$$

Useful polynomial approximations for E_1 are given in Gautschi & Cahill (1970).

3.4. Initial Conditions

In the considered picture, the source of seed photons is assumed to be localized on one face of the scattering medium. In this case the initial condition is not isotropic. As previously discussed, we consider $S_0(x, \mu) = S_0(x)U_0(\mu)$, where U_0 is axisymmetric with respect to the slab axis.

One possible application of our calculations could be the modeling of the X-ray emission from AGNs and X-ray binaries. In this case the soft photon input may take place as thermal emission from an underlying accretion disk. In the examples shown in the next

TABLE 1
BEST-FIT COEFFICIENTS OF POLYNOMIAL
APPROXIMATIONS

Coefficient	$\phi(\omega)$	$\chi(\omega)$
c_{32}	0.034	1.108
c_{31}	0.580	0.184
c_{22}	-0.367	-0.023
c_{21}	0.144	1.121
c_{12}	-16.000	-12.927
c_{11}	0.803	-5.149

§ 4, we considered the input radiation field to be a blackbody, the photon distribution reading

$$S_0(x) = \frac{1}{2\zeta(3)} \frac{x^2}{e^x - 1}, \quad (38)$$

where ζ is the Riemann zeta function. The temperature kT_{bb} is assumed to be $\ll \Theta m_e c^2$. Note that x represents the photon energy in units of kT_{bb} . We point out that the computations can be easily performed using different initial conditions $S_0(x)$, as for example a multicolor disk blackbody, or an intrinsic line profile in order to calculate Compton broadening of emission lines in hot atmospheres (Done et al., in preparation).

The angular behavior of the thermal emission from an accretion disk strongly depends on the opacity mechanisms working in the disk photosphere. A complete discussion of this problem is well beyond the purpose of the present work. We simply note that the flux versus the cosine of the viewing angle μ can be approximated as

$$U_0(\mu) = \begin{cases} \frac{6}{(3+2b)} (\mu + b\mu^2) & 0 < \mu < 1 \\ 0 & -1 < \mu < 0, \end{cases} \quad (39)$$

where the value of the parameter b is determined by the dominant opacity mechanism operating in the gas. In the case of a pure scattering atmosphere a good approximation of the intensity distribution is obtained with $b = 2$ (Chandrasekhar 1960). The radiation shows linear polarization, with a maximum degree closer to 11%. The opposite extreme case is the emission from an ideal plane-parallel radiator, with infinite opacity and scattering absent, yielding $b = 0$. In this second case the emitted radiation is unpolarized. A detailed discussion can be found in Loskutov & Sobolev (1979, 1981). It should be considered that present observations of optical polarization in AGNs seem to rule out pure scattering disk atmospheres, although the exact angular distribution of the intensity is still unknown.

Another relevant case is the angular distribution of a fluorescence line emitted by an accretion disk due to X-ray illumination (as the 6.4 keV iron line observed in Seyfert galaxies). Assuming an extended source of ionizing X-ray photons located above the disk, with some simple geometrical consideration, we find

$$U_0(\mu) = \begin{cases} 2\mu \ln \frac{1+\mu}{\mu} & 0 < \mu < 1 \\ 0 & -1 < \mu < 0. \end{cases} \quad (40)$$

The above simple expression is in good agreement with Monte Carlo results (see George & Fabian 1991, and Matt, Perola, & Piro 1991).

As we have discussed in the previous § 3.3, if the input field is not isotropic, the electron radiative losses are preferentially in the opposite direction with respect to the preferential axis of the input photon distribution. Using the initial conditions U_0 considered here, most of the power radiated in the first scattering event is emitted in downward directions, i.e., along directions defined by $\mu < 0$. In terms of the two average coefficients previously defined (eqs. [30a] and [30b]) the means that for $\mu > 0$, we find $\phi_1(\mu) < 1$ and $\xi_1(\mu) < 1$, vice versa for $\mu < 0$. Note that for $\mu > 0$ we find $\phi_2(\mu) > 1$, $\phi_3(\mu) < 1$ and so on, each successive term approaching unity. Similar behavior is expected for ξ_k . We have verified (see next § 4) that the assumption $\Theta_2 = \Theta$ introduces negligible errors in the second-fold spectrum $f_2(x, \mu)$. As previously discussed, throughout our calculations we have assumed that the second scattering could be treated as isotropic.

Finally we point out that all the calculations developed in § 3 can be easily performed for any initial conditions U_0 . Besides some obvious modifications, one should check the basic assumption that $\phi_2 \sim 1$ and $\xi_2 \sim 1$.

3.5. Escape Probability and Emitted Radiation Field

As pointed out in § 2.1, the spectrum of the n -fold scattered photons emerging from a finite medium can be written as a product of two terms, namely the probability density distribution S_n , and the escape probability p_n .

According to Nishimura et al. (1986) the escape probability can be computed iteratively as integrals of the source function of n -scattered photons $J_n^\pm(z, \mu)$ assuming isotropic scattering in the laboratory frame. As shown by equation (18), we correct the results for an anisotropic radiation field using the coefficient ϕ directly in the space of photon momenta.

Hereafter superscripts “+” and “−” specify upward ($\mu > 0$) and downward ($\mu < 0$) directions, respectively. With z we indicate the vertical coordinate along the slab. In the case of isotropy in the laboratory frame we have $J_k^+(z, \mu) = J_k^-(z, \mu) = J_k(z)$ for $k \geq 1$. Given the initial condition $J_0^+(z, \mu)$ and $J_0^-(z, \mu)$ the source function satisfies the following equation:

$$\begin{aligned} J_1(z) &= \frac{1}{2} \int_0^1 \frac{d\mu}{\mu} \left\{ \int_0^z d\zeta J_0^+(\zeta, \mu) \exp \left[-\frac{(z-\zeta)}{\mu} \right] + \int_z^\tau d\zeta J_0^-(\zeta, \mu) \exp \left[-\frac{(\zeta-z)}{\mu} \right] \right\} \\ J_k(z) &= \frac{1}{2} \int_0^\tau d\zeta J_{k-1}(\zeta) E_1(|z-\zeta|), \quad k \geq 2. \end{aligned} \quad (41)$$

In our case the initial condition is given by $J_0^+(z, \mu) = 1/2 U_0(\mu) \delta(z)$ and $J_0^-(z, \mu) = 0$.

The escape probability $p_k(\mu)$ from the upper ($z = \tau$) and the lower ($z = 0$) surfaces is given by

$$\begin{aligned} p_0(\mu) &= 2 \int_0^\tau dz J_0^+(z, \mu) \exp [-(\tau - z)/\mu] \\ p_k(\mu) &= 2 \int_0^\tau dz J_k(\tau - z) \exp (-z/\mu), \quad \mu > 0, \quad k \geq 1 \\ p_k(\mu) &= 2 \int_0^\tau dz J_k(z) \exp (-z/|\mu|), \quad \mu < 0, \quad k \geq 1. \end{aligned} \quad (42)$$

The total escape probability is normalized so as

$$\sum_{k=0}^{\infty} \int_{-1}^1 d\mu p_k(\mu) = 1. \quad (43)$$

For the sake of simplicity in the above radiative transfer equations (as well as in our Monte Carlo simulations, see next § 4) the polarization of scattered radiation is not taken into account. The polarization properties of scattering have been computed by Sunyaev & Titarchuk (1985), and Philips & Meszaros (1985) solving the radiative transfer equation for a pure scattering, plane-parallel atmosphere by separation of different scattering orders. Due to the forward-backward symmetry of the Rayleigh phase matrix used in the calculations, possible anisotropies in the energy transfer had not been taken into account and results are then fully correct for either cold electrons, or forward-backward symmetric source functions. On the other hand, since the scattering of polarized radiation shows a different angular pattern with respect to the unpolarized one, our results are formally correct only for the unpolarized component of the scattered radiation field. However, we have checked that the angular distribution of Comptonized photons obtained using the formalism of this section is in good agreement with the angular distribution of the total intensity (defined as the sum over the two possible states of polarization) obtained via formal solution of the polarized radiative transfer equation. For a better accuracy one can use the formalism depicted in Sunyaev & Titarchuk (1985) to compute the escape probabilities, using our results of § 3 to correct for anisotropy. It should be considered that in this case, since the source function of the k th fold scattered photons is angle dependent, the computational time necessary to calculate the successive $(k + 1)$ -th source function can be large.

Finally the emitted spectrum (photon flux per unit energy) can be easily obtained (see eq. [1]) from

$$f(x, \mu) = \sum_{k=0}^{\infty} f_k(x, \mu), \quad (44a)$$

where each term is given by

$$\begin{aligned} f_0(x, \mu) &= p_0(\mu) S_0(x) U_0(\mu) \\ f_1(x, \mu) &= p_1(\mu) \tilde{S}_1(x, \mu) \\ &\vdots \\ f_k(x, \mu) &= p_k(\mu) S_k(x) \\ &\vdots \end{aligned} \quad (44b)$$

4. CALCULATION OF INVERSE COMPTON SPECTRA

The method developed in the previous § 3 can be used to calculate angle-dependent IC spectra. Here we briefly describe the calculation procedure carried out for given values of the input parameters Θ , τ , and μ . The initial distributions S_0 and U_0 should be specified. The probabilities p_k can be computed using the formalism described in § 3.

In order to compute $S_{k+1}(x)$ with the initial condition $S_k(y)$, we sample S_{k+1} on the x -axis in n different points such as $x_j = c \times y_j$. The factor c does not depend on k , and choosing $c \sim 16\Theta^2 + 4\Theta + 1$, the one-dimensional grid is roughly centered on the considered S_{k+1} . If a logarithmic grid is used, only $2n - 1$ values of $\epsilon = x_h/y_i$ are possible:

$$\begin{aligned} \epsilon_{n+1-j} &= \frac{x_1}{y_j} \quad j = 1, n \\ \epsilon_{n+j-1} &= \frac{x_j}{y_1} \quad j = 2, n. \end{aligned} \quad (45)$$

It is important to note that these values do not depend on the particular scattering order, the kernel Ψ (eq. [8a]) being independent from k . This allows us to compute $2n - 1$ values of the kernel $\Psi_i \equiv \Psi(\epsilon_i)$ as defined in equation (8a) *before starting the iteration scheme*. The value of the function S_{k+1} in the point (x_h) can be obtained using the sampled values of Ψ_i between Ψ_h and Ψ_{h+n-1} , solving the integral equation (5) by means of a simple linear interpolation of the function $S_k \Psi$.

The same algorithm is adequate to compute the first anisotropic distribution S_1 using the approximated formula (29). The anisotropy coefficients $\phi(\mu, \mu')$ and $\xi(\mu, \mu')$ are computed over a grid of μ' values. We compute $2n - 1$ values of the kernel Ψ using the results of § 3.1, and with the procedure described above we find a set of $\tilde{S}_1(x, \mu, \mu')$ distributions. The anisotropic distribution $\tilde{S}_1(x, \mu)$ can be obtained by means of a linear interpolation between μ' sampling points using equation (25). To compute the same

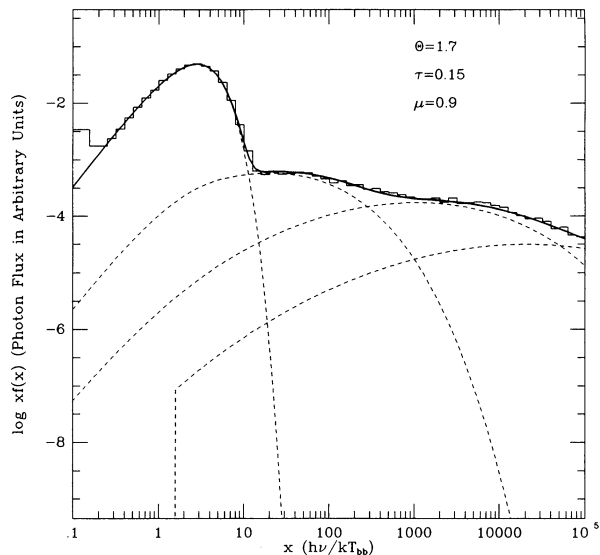


FIG. 2a

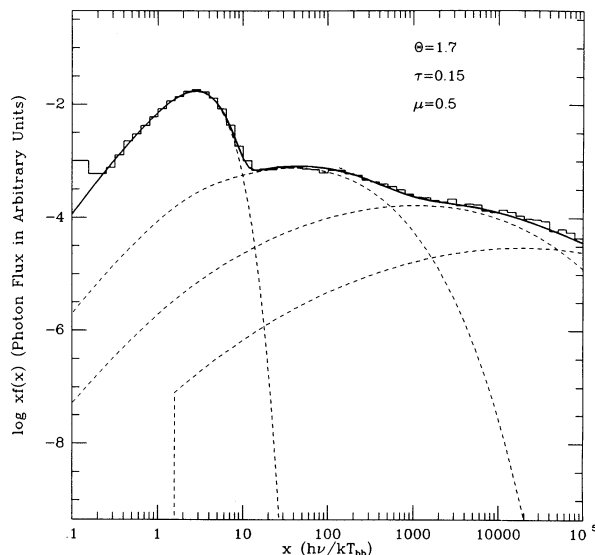


FIG. 2b

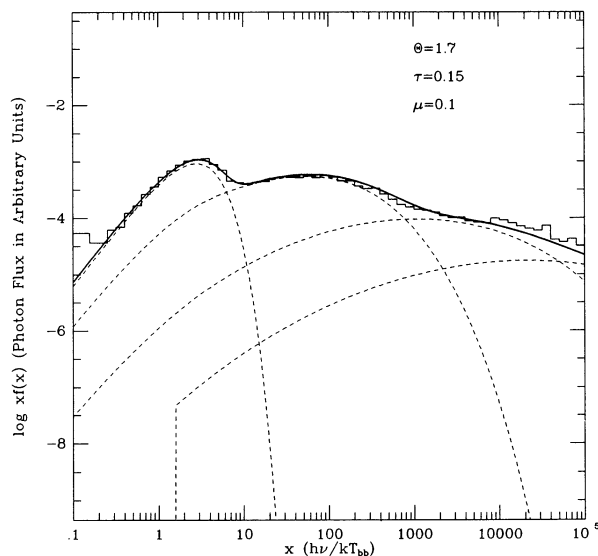


FIG. 2c

FIG. 2.—Computed IC spectrum compared with Monte Carlo simulations. The total spectrum (solid line) is shown together with the single scattering ones (dashed lines). The step function shows the result of numerical simulations. The soft photon input is a blackbody, with emissivity according to a limb-darkening law with $b = 2$ (see eq. [39]). The input values of the parameters of the hot plasma are $\Theta = 1.7$ and $\tau = 0.15$, which yield a Compton parameter $y \approx 8$. The cosine of the viewing angle is $\mu = 0.9$ for panel (a), $\mu = 0.5$ for panel (b), and $\mu = 0.1$ for panel (c). Monte Carlo spectra are averaged in cosine intervals of width $\Delta\mu = 0.2$ around the labeled value of μ . The photon energy is in unit of kT_{bb} where T_{bb} is the blackbody temperature. The low-energy cutoff of f_3 and the anomalous behavior of the first two energy bins of the simulated spectra are due to numerical approximations.

model for a different viewing angle, we can recalculate only $\tilde{S}_1$ and the probabilities p_k , since the distributions S_k , for $k \geq 2$, are independent on μ .

In Figures 2, 3, and 4 some examples of the calculation are shown. We have plotted the photon flux [defined as $xf(x)$] in order to compare results with the output of Monte Carlo simulations. A limb-darkening law ($b = 2$ in eq. [39]) is used. As can be seen the agreement is sufficiently good for a wide range of input parameters. As discussed in § 3.2, the errors due to the approximated treatment of f_1 are more evident for high inclination angles. From panel (c) of Figures 2, 3, and 4 it can be seen how the flux relative to f_1 is slightly overestimated for grazing angles. The Monte Carlo simulations are performed using $kT_{bb}/m_e c^2 = 10^{-5}$. The analytical results are then formally correct if $x \ll 10^5 \times \min(\Theta, 1/\langle\gamma\rangle)$. In Figures 5 and 6, analytical spectra for different viewing angles are compared in the same plot, showing how the anisotropic emission in the first scattering region changes. Note that the flux f_1 for $\mu < 0$ exceeds the flux f_1 as seen by a possible observer located at $\mu > 0$.

If an isotropic intensity rather than a limb darkening law is used, for the same values of the parameters, the resulting hard component is somewhat more luminous, while the soft one is somewhat less luminous, because of the decreasing of the direct escape probability. The shape of the Comptonized component remains practically unchanged, except slight differences in the anisotropic distribution f_1 .

The IC spectrum is formed by the sum of the single scattering spectra, each of them reflecting in shape (in some way) the energy distribution of the electrons. In the case of thermal electrons the total spectrum can be approximated by a power-law $f \propto x^{-\alpha-1}$ if the temperature is sufficiently low and the opacity high so that single spectra are “tightly packed” on the frequency axis. Power-law

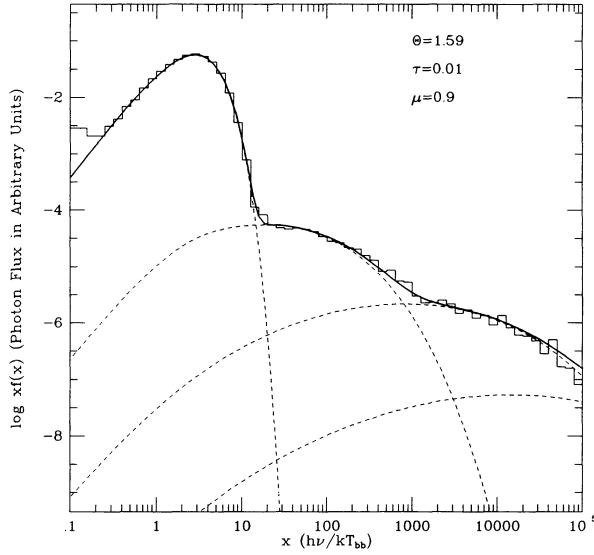


FIG. 3a

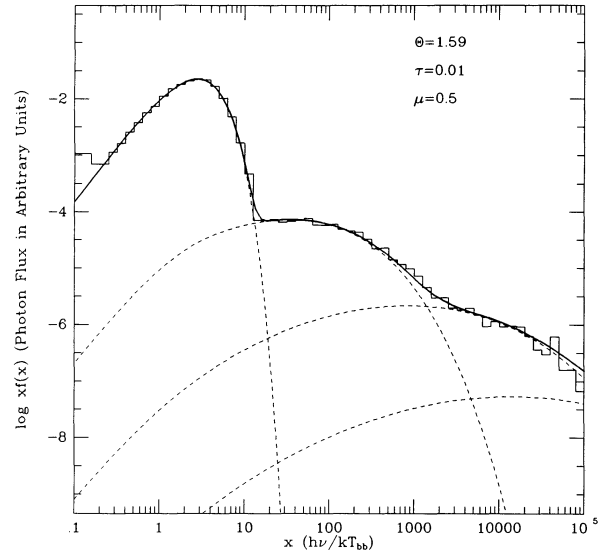


FIG. 3b

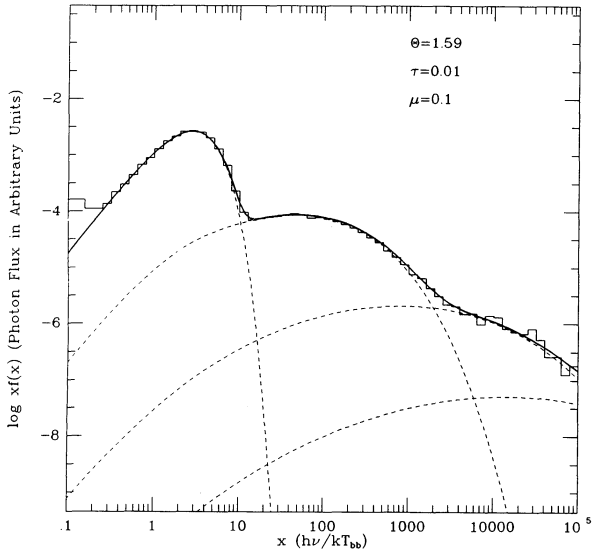


FIG. 3c

FIG. 3.—Same as Fig. 2, but $\Theta = 1.59$ and $\tau = 0.01$. The resulting Compton parameter is $y \simeq 0.5$.

shape occurs up to energies for which the electron recoil is negligible. The power-law approximation is not tenable when the electron temperature is high and the opacity low. In this case features of single spectra are clearly visible (see Fig. 3).

In general the spectra index α depends on τ and Θ . If the Compton emission from a slab is integrated over all directions, good agreement with numerical simulations for $\Theta > 0.05$ and $\tau < 1$ are obtained with

$$\alpha = [-\ln(\tau) + 2/(3 + \Theta)] / [\ln(12\Theta^2 + 25\Theta)] . \quad (46a)$$

Taking into account the angular dependence of the radiated photons, the main feature found with our calculations is the deficiency of power in the first scattering distribution for low inclination angles. This fact causes evident spectral distortions. In the region dominated by f_1 , the spectrum is considerably flatter than the average one. We have empirically found that a broken power law can qualitatively fit the spectrum. Defining A_1 as the mean energy gain per scattering, for $\chi_1 A_1 < x < A_1^3$ the spectral index is given by

$$\alpha_\mu = [-\ln(\tau/\phi_1) + 2/(3 + \Theta)] / \ln[(12\Theta^2 + 25\Theta)/\chi_1] , \quad (46b)$$

while for $x > A_1^3$ good fits are obtained with the averaged formula (46a). For $\mu > 0$ α_μ is lower than α , vice versa for $\mu < 0$. The flatter spectra are seen at low inclination angles.

5. CONCLUSIONS

We have developed a formalism in order to treat anisotropic IC in plane parallel geometry. In our calculations a very simple kernel for the scattering has been adopted, with the limitation of $h\nu \ll kT$, and the polarization in the radiative transfer has been

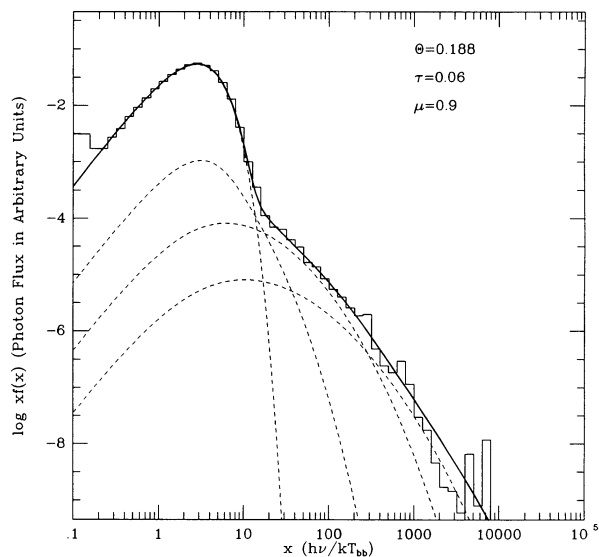


FIG. 4a

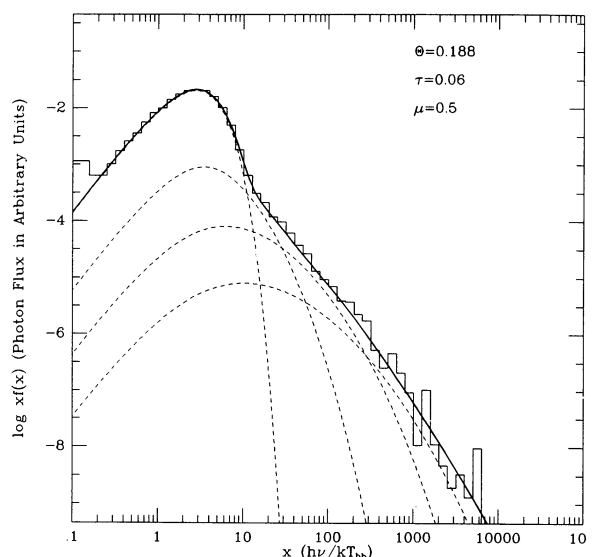


FIG. 4b

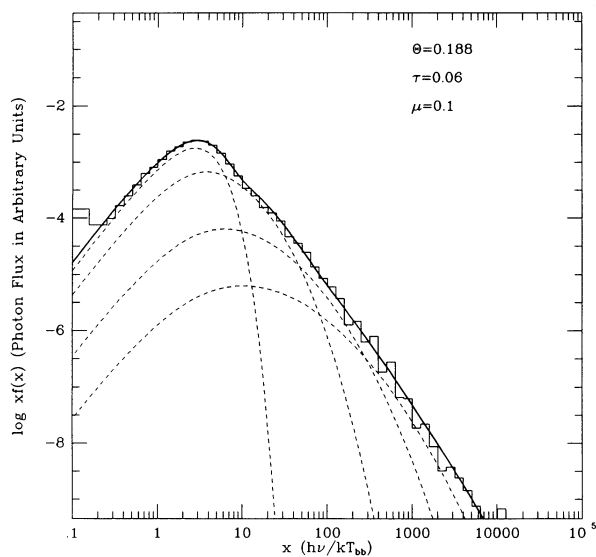


FIG. 4c

FIG. 4.—Same as Fig. 2, but $\Theta = 0.188$ and $\tau = 0.06$. The resulting Compton parameter is $y \approx 0.08$.

neglected. Furthermore the anisotropy is taken into account for the first scattering event only, where this effect is relevant. All the above approximations allow us to compute numerically angle dependent IC spectra with a good accuracy and a modest computational effort. The method can be easily used in the current X-ray data analysis packages. The X-ray spectrum of the black hole candidate Cyg X-1 has been fitted using the present method (Haardt et al. 1993).

We would like to stress that *all* the approximations can be avoided, by modifications of the depicted procedure. The full relativistic kernel (Jones 1968; see Coppi & Blandford 1990 for corrections) can be used in equation (8a), as well as the radiative transfer equation of polarized radiation (Sunyaev & Titarchuk 1985) in § 3.5. Finally § 3.2 can be modified and the treatment extended to higher scattering orders.

Our calculations show that IC scattering of an anisotropic soft photon input leads to an anisotropic distribution of hard radiation. Most of the power in the first scattering event is emitted back along the direction of the input seed photons. In the case of multiple scattering an anisotropic soft photon input leads to a strong dependence of the spectral shape on direction. For low inclination angles the spectrum in the region relative to the first scattering is significantly flatter than the spectrum at higher energies. The effect is more evident for high temperatures and low opacities. The IC spectrum could be roughly approximated by a broken power law.

Our results have particular interest in modeling the X-ray emission from sources where an accretion disk is possibly present, such as AGNs and X-ray binaries. We note that X-ray spectral observations of Seyfert galaxies have revealed the presence of several components (Pounds et al. 1990; Matsuoka et al. 1990; Piro, Yamauchi, & Matsuoka 1990). Added to an underlying power law, an emission line at 6.4 keV and a high-energy hump, as predicted by Lightman & White (1988) and Guilbert & Rees (1988), were found to be common. The last two features may result from reflection of the downdriven Compton flux on the underlying dense layers

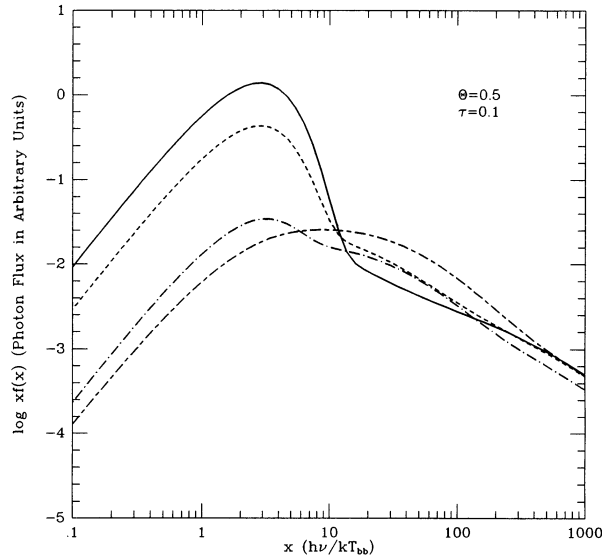


FIG. 5

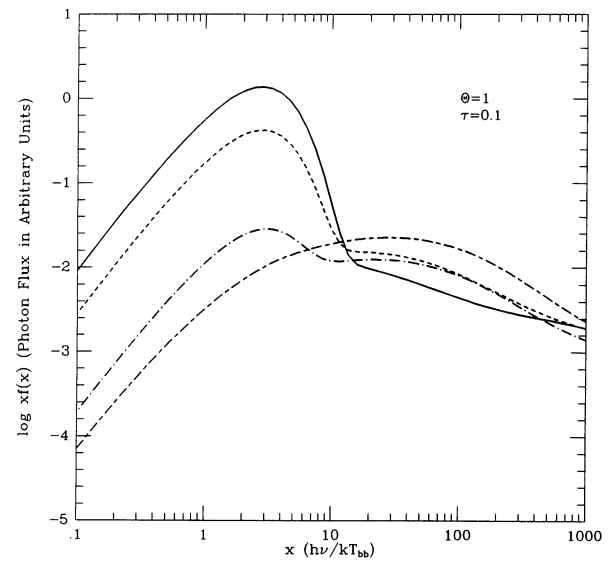


FIG. 6

FIG. 5.—Analytical IC spectrum for $\Theta = 0.5$ and $\tau = 0.1$. Solid line refers to $\mu = 1$, dashed line to $\mu = 0.5$, dash-dotted line to $\mu = 0.1$, and long-short dashed line to $\mu = -0.5$.

FIG. 6.—Same as Fig. 5, but $\Theta = 1$ and $\tau = 0.1$

responsible for the thermal emission. Recently there is evidence for similar spectral distributions in Galactic black hole candidates (Tanaka 1991; Ebisawa et al. 1991; Done et al. 1992; Haardt et al. 1993). Our calculation shows that, because of the anisotropic IC emission, the enhancing of the down-driven X-ray flux produces a flux irradiating the dense layer considerably greater than that observed directly. This may have important consequences for the reflection model proposed for AGNs and weakly magnetized X-ray binaries (George & Fabian 1991; Matt, Perola, & Piro 1991). The model could be improved considering the angular behavior of the impinging X-rays.

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APPENDIX

AVERAGED ANISOTROPY COEFFICIENTS

The solution of integral (28b) can be written as

$$\xi(\mu, \mu') = \sum_{i=0}^6 h_i(\mu, \mu') Z_i, \quad (\text{A1})$$

where

$$\begin{aligned} h_0 &= 1 \\ h_1 &= \mu\mu' \\ h_2 &= \frac{1}{2}(1 - \mu^2 - \mu'^2 + 3\mu^2\mu'^2) \\ h_3 &= \frac{1}{2}\mu\mu'(3 - 3\mu^2 - 3\mu'^2 + 5\mu^2\mu'^2) \\ h_4 &= \frac{1}{8}(3 - 6\mu^2 + 3\mu^4 - 6\mu'^2 + 36\mu^2\mu'^2 - 30\mu^4\mu'^2 + 3\mu'^4 - 30\mu^2\mu'^4 + 35\mu^4\mu'^4) \\ h_5 &= \frac{1}{8}\mu\mu'(15 - 30\mu^2 + 15\mu^4 - 30\mu'^2 + 100\mu^2\mu'^2 - 70\mu^4\mu'^2 + 15\mu'^4 - 70\mu^2\mu'^4 + 63\mu^4\mu'^4) \\ h_6 &= \frac{1}{16}(5 - 15\mu^2 + 15\mu^4 - 5\mu^6 - 15\mu'^2 + 135\mu^2\mu'^2 - 225\mu^4\mu'^2 + 105\mu^6\mu'^2 + 15\mu'^4 \\ &\quad - 225\mu^2\mu'^4 + 525\mu^4\mu'^4 - 315\mu^6\mu'^4 - 5\mu'^6 + 105\mu^2\mu'^6 - 315\mu^4\mu'^6 + 231\mu^6\mu'^6). \end{aligned} \quad (\text{A2})$$

Note that the above expression is symmetric in μ and μ' , and that the integral over the total solid angle is equal to 1. The integral in equation (28a) is analogous, reading

$$\phi(\mu, \mu') = \sum_{i=0}^3 h_i(\mu, \mu') Q_i. \quad (\text{A3})$$

Assuming the initial condition given by equation (39), the integral in equation (30b) reads

$$\xi_1(\mu) = \frac{1}{u_0} \sum_{i=0}^6 u_i v_i(\mu), \quad (\text{A4})$$

where

$$u_i = \frac{1}{i+2} + \frac{b}{i+3} - E_{i+3}(\tau) - bE_{i+4}(\tau) \quad (\text{A5})$$

and

$$\begin{aligned} v_0 &= Z_0 + \frac{Z_2}{2} (1 - \mu^2) + \frac{Z_4}{8} (3 - 6\mu^2 + 3\mu^4) + \frac{Z_6}{16} (5 - 15\mu^2 + 15\mu^4 - 5\mu^6) \\ v_1 &= Z_1 \mu + \frac{Z_3}{2} (3\mu - 3\mu^3) + \frac{Z_5}{8} (15\mu - 30\mu^3 + 15\mu^5) \\ v_2 &= \frac{Z_2}{2} (-1 + 3\mu^2) + \frac{Z_4}{8} (-6 + 36\mu^2 - 30\mu^4) + \frac{Z_6}{16} (-15 + 135\mu^2 - 225\mu^4 + 105\mu^6) \\ v_3 &= \frac{Z_3}{2} (-3\mu + 5\mu^3) + \frac{Z_5}{8} (-30\mu + 100\mu^3 - 70\mu^5) \\ v_4 &= \frac{Z_4}{8} (3 - 30\mu^2 + 35\mu^4) + \frac{Z_6}{16} (15 - 225\mu^2 + 525\mu^4 - 315\mu^6) \\ v_5 &= \frac{Z_5}{8} (15\mu - 70\mu^3 + 63\mu^5) \\ v_6 &= \frac{Z_6}{16} (-5 + 105\mu^2 - 315\mu^4 + 231\mu^6). \end{aligned} \quad (\text{A6})$$

The integral in equation (30a) is analogous, reading

$$\phi_1(\mu) = \frac{1}{u_0} \sum_{i=0}^3 u_i v_i(\mu), \quad (\text{A7})$$

where in equation (A6) the coefficients Z_j for $j \leq 3$ must be replaced with Q_j , while for $j \geq 4$ with 0. The exponential integral functions E_i are defined in § 3.3, equation (37a).

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